Unforced Navier–Stokes

Compact symmetric stresses, exact coefficient margins, and formal verification on the three-torus

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Abstract

The absence of external forcing makes the three-dimensional Navier–Stokes regularity problem substantially more restrictive than a construction with prescribed smooth forcing. We study exact mathematical prerequisites for a candidate unforced finite-time singularity on the periodic three-torus. An explicit compact symmetric-divergence inverse is proved for smooth axisymmetric meridional sources supported in an annulus, with vanishing total axial force as the necessary and sufficient compatibility condition. A separate two-diagonal pressure-neutral construction exhibits a sharp radial sign obstruction. Finite seven-ray stress algebra yields the exact coefficient margin 15971/122900 and sharp componentwise amplification factor 14. Lean checks the finite cone algebra, radial pressure-gauge derivative identity, full physical residual expansion, and conditional retention of terminal speed. These results do not yet construct the source-matched, scale-sharp, positive-cone-compatible compact stress or a genuine zero-force singular solution. The global unforced existence question remains open.

1 The problem and the mathematical objective

We work at viscosity one on the unit three-torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For a smooth velocity $u(t,x) \in \mathbb{R}^3$ and pressure $p(t,x) \in \mathbb{R}$, the unforced incompressible equations are

$$\partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p = 0, \qquad \operatorname{div} u = 0. \tag{1}$$

The specific target is to find nonzero smooth divergence-free periodic initial data and a solution smooth for $0 \le t < 1$, of finite kinetic energy on a fundamental cell, whose speed becomes unbounded as $t \uparrow 1$. The source-free condition in (1) is essential: a verified construction with a nonzero smooth force does not establish this target. The Clay problem statement makes this distinction explicit [1]. The published OpenAI Lean formalization states Navier–Stokes breakdown with smooth forcing; its unforced finite-time result concerns Euler rather than Navier–Stokes [2].

Our approach isolates small results that are meaningful independently of the unresolved global theorem. First, a source must admit a compact symmetric-divergence correction. Second, if that correction must remain in a prescribed positive stress cone, its components need quantitative control relative to a density envelope. Third, all terms in the actual nonlinear physical residual must cancel with a single periodic pressure. Finally, a singular leading trace can survive only if the correction is provably subleading. Sections 2–6 establish exact parts of this chain.

The construction of compact symmetric anti-divergence operators under force and torque compatibility is not new; see, for example, the support-preserving operator and its explicit moment correction in [3, §3.6 and Appendix A]. The direct axisymmetric formula below is offered as an elementary source-specific coordinate realization, not as a priority claim for compact symmetric anti-divergence. The new obstacle for the present programme is the

additional shrinking-support, weighted-seven-ray positivity requirement, which is not supplied by general existence of a compact tensor.

2 A compact symmetric inverse in an annular region

Let $0 < a < b$ and $c < d$, and write $\Omega = (a, b) \times (c, d)$ in meridional cylindrical coordinates (r, z). All functions in this section are independent of the azimuthal angle θ. The orthonormal cylindrical frame is (e_r, e_θ, e_z). For a symmetric tensor with components

$$S = K\, e_r \otimes e_r + L\, e_\theta \otimes e_\theta + M(e_r \otimes e_z + e_z \otimes e_r) + C\, e_z \otimes e_z, \tag{2}$$

standard cylindrical differentiation gives

$$(\operatorname{div} S)_r = \partial_r K + \frac{K - L}{r} + \partial_z M, \tag{3}$$

$$(\operatorname{div} S)_z = \partial_r M + \frac{M}{r} + \partial_z C, \tag{4}$$

$$(\operatorname{div} S)_\theta = 0.$$

The $1/r$ terms are geometric, not lower-order terms that may be discarded near a shrinking cylindrical support.

Theorem 2.1 (Explicit compact meridional inversion)**.** *Let $f_r, f_z \in C_c^\infty(\Omega)$, and extend $f = f_r e_r + f_z e_z$ by zero to $\mathbb{R}^3$ in cylindrical coordinates. There is a symmetric $S \in C_c^\infty(\mathbb{R}^3; \mathrm{Sym}_3)$, axisymmetric and supported in the same open annular cylinder, with $\operatorname{div} S = f$, if and only if*

$$\int_a^b \int_c^d r f_z(r, z)\, \mathrm{d}z\, \mathrm{d}r = 0. \tag{5}$$

The construction uses no nonlocal spatial projection, but it asserts no positive-semidefinite or weighted-cone bound on S.

Proof. For necessity, integrate the axial component of $\operatorname{div} S = f$ against the cylindrical Jacobian r. Compact support and (4) give

$$\int_a^b \int_c^d r f_z\, \mathrm{d}z\, \mathrm{d}r = \int_a^b \int_c^d \left[\partial_r(rM) + r\partial_z C\right] \mathrm{d}z\, \mathrm{d}r = 0.$$

For sufficiency, define the axial source marginal and its radial primitive by

$$J(r) = \int_c^d f_z(r, z)\, \mathrm{d}z, \qquad H(r) = \frac{1}{r} \int_a^r s J(s)\, \mathrm{d}s. \tag{6}$$

Since $r > 0$ on $[a, b]$, H is smooth. The vanishing moment (5) makes H identically zero outside a compact subinterval of (a, b). Choose any $\eta \in C_c^\infty((c, d))$ normalized by $\int_c^d \eta(z)\, \mathrm{d}z = 1$, and set

$$M(r, z) = H(r)\eta(z). \tag{7}$$

The identity $(rH)' = rJ$ implies

$$\int_c^d \left(f_z - \partial_r M - \frac{M}{r}\right) \mathrm{d}z = J - H' - \frac{H}{r} = 0.$$

Consequently the axial primitive

$$C(r, z) = \int_c^z \left[f_z(r, \zeta) - \partial_r M(r, \zeta) - \frac{M(r, \zeta)}{r}\right] \mathrm{d}\zeta \tag{8}$$

vanishes both below and above the axial support; it is smooth and compactly supported in Ω. Choose any $K \in C_c^\infty(\Omega)$, including $K = 0$, and define

$$L = K + r(\partial_r K + \partial_z M - f_r). \tag{9}$$

Each component of (2) is smooth and compactly supported away from the axis. Substitution in (3) gives

$$\partial_r K + \frac{K - L}{r} + \partial_z M = \partial_r K - (\partial_r K + \partial_z M - f_r) + \partial_z M = f_r,$$

and (8) gives $\partial_r M + M/r + \partial_z C = f_z$. Thus $\operatorname{div} S = f$ as claimed. □

Remark 2.2 (Scope of the construction)**.** The theorem concerns a meridional source with $f_\theta = 0$ and support strictly separated from $r = 0$. An azimuthal source introduces a different angular-momentum compatibility condition, and extending the construction across the axis requires separate regularity conditions on the cylindrical components. The tensor in the theorem may be indefinite, and its size can deteriorate when the support shrinks. None of these omissions is harmless for the weighted H8 attachment problem.

3 Pressure-neutral gauges and a sign obstruction

The mixed and hoop degrees of freedom in Theorem 2.1 are not optional decorations. A restricted diagonal ansatz exhibits a genuine obstruction to maintaining one sign.

Proposition 3.1 (Two-diagonal gauge)**.** *Let $F \in C_c^\infty(\Omega)$, put $K = rF_r$ and $C = F + rF_r$, and take $S = Ke_r \otimes e_r + Ce_z \otimes e_z$. Then*

$$\operatorname{div} S = \nabla C. \tag{10}$$

In particular its divergence is a pressure gradient, so it cannot by itself represent a prescribed source with a nonzero solenoidal part.

Proof. The radial divergence is $K_r + K/r$. Since $K = rF_r$, $C_r = F_r + K_r = K/r + K_r$. The axial divergence is C_z, which proves (10). □

Proposition 3.2 (Restricted radial sign condition)**.** *If K and C are compactly supported in $r \in (a, b)$ at a fixed z and obey $C_r = K_r + K/r$, then*

$$\int_a^b K(r, z)/r \, \mathrm{d}r = 0. \tag{11}$$

Consequently, $K \geq 0$ forces $K \equiv 0$ for this restricted two-diagonal class.

Proof. Integrate the derivative identity between a and b; the endpoint terms of K and C vanish by compact support. Since $r > 0$, continuity and (11) imply that a nonnegative K is identically zero. □

This conclusion does not exclude a sign-changing correction around a positive background tensor or a compact construction with nonzero hoop and mixed components. Indeed, Theorem 2.1 explicitly uses both of those components. The more difficult question is whether one can choose them while retaining a prescribed strict coefficient cone with uniform shrinking-support estimates.

4 The exact seven-ray coefficient margin

The finite cone appearing in the current H8 source has seven rays. With aperture $t = 1/4$ they are

$$\begin{aligned} v_1 &= (1,0,0), \quad v_2 = (1,t,0), \quad v_3 = (1,-t,0), \\ v_4 &= (1,0,t), \quad v_5 = (1,0,-t), \\ v_6 &= (1,t,t), \quad v_7 = (1,t,-t). \end{aligned} \tag{12}$$

Their normalized positive coefficient centre is $c_i^{(0)} = 236/1229$ for $i = 3$ and $172/1229$ for every other index. The raw source covariance and its trace-free rank-one identity are established by exact finite arithmetic and in `UnforcedPatching.SevenRayCenter.lean`.

Write $(A, B, C, D) \in \mathbb{R}^4$ for the four trace-free stress coordinates. The resulting coefficient perturbations, in ray order, are

$$\delta c = \left(-\tfrac{3}{2}D,\ 2C - 4D,\ -2C - 4D,\ 2A - 8B + 4D,\ -2A + 8B + 4D,\ 8B,\ -8B\right). \tag{13}$$

Proposition 4.1 (Sharp componentwise cone estimate)**.** *For $m = \max(|A|, |B|, |C|, |D|)$ the map (13) satisfies*

$$\max_{1 \le i \le 7} |\delta c_i| \le 14m, \tag{14}$$

and the number 14 is optimal. If $\rho \ge 0$ and

$$m \le \frac{15971}{1720600}\rho, \tag{15}$$

then for every ray $c_i^{(0)}\rho + \delta c_i \ge \rho/100$.

Proof. The absolute row sums in (13) are respectively 3/2, 6, 6, 14, 14, 8, 8; hence the operator norm from $\ell^\infty(\mathbb{R}^4)$ to $\ell^\infty(\mathbb{R}^7)$ is at most 14. At $(A, B, C, D) = (1, -1, 0, 1)$, the fourth coordinate equals 14, so the constant is sharp. Since

$$\frac{172}{1229} - \frac{1}{100} = \frac{15971}{122900},$$

inequality (15) implies $|\delta c_i| \le 14m \le (15971/122900)\rho$. As $c_i^{(0)} \ge 172/1229$, the claimed floor follows. □

The seven-ray claim is a finite linear-algebra statement. It provides the exact smallness target for a potential compact correction but gives no estimate for the particular tensor delivered by Theorem 2.1. In particular, no implication from unweighted compact solvability to (15) is available.

5 The physical residual and terminal speed

Define the viscosity-one residual

$$\mathcal{N}(u, p) = u_t + (u \cdot \nabla)u - \Delta u + \nabla p. \tag{16}$$

For smooth (u, p) and (w, q), direct expansion yields

$$\mathcal{N}(u + w, p + q) = \mathcal{N}(u, p) + \mathcal{N}(w, q) + (u \cdot \nabla)w + (w \cdot \nabla)u. \tag{17}$$

Both nonlinear cross terms are essential. Equation (17), with explicit differentiability assumptions in the upstream physical definitions, is checked in Lean by the declaration `UnforcedPatching.navierStokesResidual_add`.

Proposition 5.1 (Conditional terminal retention)**.** *Suppose that u has arbitrarily large speed at points with $t \uparrow 1$. If a correction w satisfies*

$$\|w(t,x)\| \le \alpha \|u(t,x)\| + C_0, \qquad 0 \le \alpha < 1, \quad C_0 \ge 0, \tag{18}$$

uniformly on $0 < t < 1$ and $x \in \mathbb{T}^3$, then $u + w$ also has arbitrarily large speed arbitrarily close to time one.

Proof. The reverse triangle inequality and (18) give

$$\|u + w\| \ge \|u\| - \|w\| \ge (1 - \alpha)\|u\| - C_0.$$

Given any target speed $M > 0$ and any terminal neighbourhood, choose a point therein where $\|u\| > (M + C_0)/(1 - \alpha)$. Then $\|u + w\| > M$ at that point. □

This last implication is kernel-checked by the Lean theorem

`UnforcedPatching.speedUnbounded_add_subleading`

It is deliberately conditional: no field u meeting the singularity premise, and no actual PDE correction w meeting (18), has been supplied by these lemmas.

6 What the formal development proves

The Lean project is pinned to Lean 4.34.0-rc2 and a fixed Mathlib revision. The checked physical interface states smoothness, three coordinate periods, zero forcing, nonzero initial data, finite fundamental-cell kinetic energy and endpoint speed divergence. Its existential proposition is named `UnforcedPatching.unforcedTargetStatement`; there is no proof term inhabiting that proposition.

The precise correspondence between the paper and the compiled `UnforcedPatching` declarations is as follows. In the third column, "missing" refers to mathematical constructions, not to unresolved Lean goals inside the named theorems.

Lean module	Checked scope and the missing construction
`Definitions`	Physical candidate predicates; no candidate witness.
`Residual`, `Correction`	Full residual addition and conditional cancellation; no constructed correcting field.
`Periodicity`, `PressureGauge`	Sum and time-only gauge identities; no periodic singular solution.
`ConeBudget`, `ConeOperator`, `SevenRayCenter`	Exact rational centre and sharp factor 14; no source-matched weighted positive stress.
`CompactGauge`	One-variable derivative; no cylindrical tensor or support theorem.
`Retention`	Subleading corrections preserve terminal speed; neither required field constructed.

In particular, the residual-addition, correction-closure, seven-ray-floor, and terminal-retention declarations are all implications with explicit hypotheses. Their conjunction does not yield an inhabitant of `unforcedTargetStatement`. The printed axioms for the selected smoke-tested declarations are the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

These compiled declarations do not include the cylindrical tensor calculus (3)–(4), the integral construction (6)–(9), compact-support theorems or weighted H8 bounds. Theorem 2.1 therefore has the complete analytic proof given above but is *not yet Lean-formalized.*

An independent exact-arithmetic test evaluates a nontrivial annular polynomial instance of the radial compatibility identity. For the uncompensated purely radial version, the curl obstruction at $(r, z) = (5/4, 1/2)$ equals $-5103/32768$ exactly; the test explicitly rejects the pressure-neutral interpretation in this case. The polynomial is used only for interior differentiation, not as an infinitely smooth zero-extended cutoff.

The source code, machine-readable G0–G8 ledger, smoke tests and reproduction commands accompany the public integration branch [6]. A local pinned build has passed ten smoke modules; at the time of preparing this edition, the corresponding GitHub-hosted Actions checks had not furnished a successful receipt for the latest commit. Local compilation does not replace hosted CI, semantic theorem auditing or independent scrutiny of the paper proof.

7 Remaining analytic requirements

The compact meridional inverse of Theorem 2.1 is a properly restricted solvability result. It does not resolve the weighted attachment problem because the desired source may have an azimuthal component, a noncompact Leray-projected tail, nontrivial pressure coupling, or incompatible moments. Even when compact inversion is possible, the output may leave the required seven-ray positive cone, especially as the support shrinks.

A complete unforced singularity construction must still provide a single authenticated local profile, source-matched smooth shell data, a scale-sharp weighted symmetric inverse respecting the coefficient margin (15), a periodic nonlinear correction with a proved right-inverse/contraction scheme, the full physical zero-residual identity, and terminal retention for that actual correction. Each claim requires a further source-to-Lean semantic audit and a compiled theorem using the genuine PDE definitions.

Thus the present mathematical result is an exact compact stress construction and several independently checked necessary or conditional ingredients. The existence or nonexistence of a finite-time singular solution to (1) with smooth nonzero periodic initial data is *not decided by this manuscript*.

References

[1] C. Fefferman, *Existence and Smoothness of the Navier–Stokes Equation*, Clay Mathematics Institute Millennium Prize Problem statement, https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf.

[2] OpenAI, *Finite time blowup for Navier–Stokes* (2026), and the associated formalization repository, https://github.com/openai/NavierStokesAndEuler.

[3] X. Zhao, *Compactly Supported Finite-Energy Stationary Solutions of the Three-Dimensional Navier–Stokes Equations* (2026), arXiv:2608.17383, https://arxiv.org/abs/2608.17383.

[4] J. Leray, *Sur le mouvement d'un liquide visqueux emplissant l'espace*, Acta Mathematica **63** (1934), 193–248.

[5] L. Caffarelli, R. Kohn, and L. Nirenberg, *Partial regularity of suitable weak solutions of the Navier–Stokes equations*, Communications on Pure and Applied Mathematics **35** (1982), 771–831.

[6] T. Newton, *Unforced Navier–Stokes Lean integration branch*, https://github.com/tonynewton79-web/unforced-navier-stokes-lean/pull/3.